FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – III
(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**D**	31.	**B**	61.	**B**
2.	**B**	32.	**C**	62.	**C**
3.	**A**	33.	**C**	63.	**A**
4.	**B**	34.	**B**	64.	**B**
5.	**C**	35.	**B**	65.	**A**
6.	**D**	36.	**C**	66.	**A**
7.	**C**	37.	**D**	67.	**D**
8.	**D**	38.	**C**	68.	**A**
9.	**D**	39.	**A**	69.	**D**
10.	**D**	40.	**B**	70.	**D**
11.	**C**	41.	**C**	71.	**B**
12.	**C**	42.	**C**	72.	**A**
13.	**D**	43.	**D**	73.	**C**
14.	**C**	44.	**C**	74.	**A**
15.	**B**	45.	**B**	75.	**D**
16.	**B**	46.	**A**	76.	**D**
17.	**A**	47.	**B**	77.	**A**
18.	**D**	48.	**C**	78.	**A**
19.	**D**	49.	**C**	79.	**C**
20.	**D**	50.	**D**	80.	**B**
21.	**B**	51.	**B**	81.	**C**
22.	**A**	52.	**B**	82.	**B**
23.	**D**	53.	**C**	83.	**C**
24.	**B**	54.	**C**	84.	**C**
25.	**B**	55.	**D**	85.	**C**
26.	**B**	56.	**A**	86.	**D**
27.	**C**	57.	**B**	87.	**C**
28.	**D**	58.	**D**	88.	**B**
29.	**D**	59.	**D**	89.	**D**
30.	**A**	60.	**D**	90.	**B**

Physics

PART – I

SECTION – A

1. $\delta R/R = 2\delta u/u + 2\cot 2\theta\, \delta\theta$

2. If s be their separation at time t then,
$s^2 = (0.4-80t)^2 + (0.6-60t)^2$
where t is in hour, s in km. Minimise s using calculus.

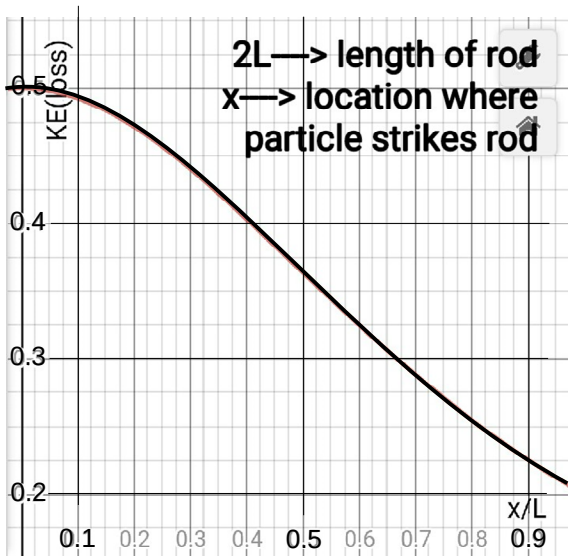
3. The horizontal acceleration of Block 1 is always greater than that of block 2.

4. The normal reaction vanishes when the person loses contact with the surface:
$mv^2/r = mg\cos\theta$;
Conservation of energy gives:
$1/2\, mv^2 + mgr\cos\theta = 1/2\, mu^2 + mgr$; where $u^2 = 0.5gr$

5. The droplets fall a distance $\frac{1}{2}gt^2$ in time t, and the number of the droplets and hence their mass is proportional to dt. Computing the CM of the droplets using the definition, we get the result.

6. For an equilateral triangle of side a, moment of inertia is $\frac{ma^2}{12}$ about an axis passing through its CM and perpendicular to plane - the result of the integration is similar to that of a solid cone along its axis: a factor of 3/5.
$I = \frac{3}{5}\left(\frac{ma^2}{12}\right)$

7. The maximum loss occurs when the collision is "head-on" and the minimum is when it collides at an end. Assume that the particle strikes the rod (of length 2L) perpendicularly at a distance x from its centre. Apply conservation of momentum and angular momentum to calculate the final velocity of the rod and also its angular velocity. The fractional loss in KE = $1/[3(x/L)^2 + 2]$



8. We take $\theta = (10°)\cos(\pi t)$; calculate t when $\theta = -5°$. This gives t = 2/3 (s). The period is 2 times this.

9. Use Kepler's Law of Periods.

10. If m be the mass of water displaced, the buoyant force is mg, while the mass of the extra water that has to be pushed "out of the way" is effectively m/2.

11. Taking torques about the hinge, we get:
In the 1st case, mgL/2 =p AL/2
In the second case, mgL/4 = p' AL/2

12. When the wheel gets just lifted $\tau = mg(3r/5)$. But when a torque of 2τ is applied, we can write: $mg(3r/5) = I\alpha$,
where $I = 2mr^2$.
The horizontal acceleration is $r\alpha(4/5)$.

13. The final pressure is $p = \frac{101}{2}$kPa ; $kx = p\pi r^2$ where x = 0.2, the compression in the spring and r = 0.05 the radius of cylinder.

14. With reservoirs at 273 K, 173 K:
$\frac{W}{Q} = \frac{273-173}{273} = 0.37$
With reservoirs at 373 K, 173 K:
$\frac{W'}{Q} = \frac{373-173}{373} = 0.54$
$W' - W = (0.54 - 0.37)Q = 0.17\ Q$

15. $T = \frac{2w}{v}$; $f = \frac{1}{T}$

16. $\frac{3 \text{ volt}}{(3+5)\text{volt}} = e^{-t/\tau}$; $\tau = RC = 6$ s
$\therefore\ t \approx 6$ s.

17. Tension, $T = \frac{3m_1m_2}{2(m_1+m_2)}g$; $v = \sqrt{\frac{T}{\mu}}$; $\frac{\delta v}{v} = \frac{1}{2}\frac{\delta T}{T} = \frac{1}{3}\frac{\delta m_2}{m_2}$ (where, $m_1 \approx 2m_2$)

18. The change in flux $\Delta\phi = \frac{\pi}{2}r^2B$, where r = 0.2 m
Average emf = $\frac{\Delta\phi}{\pi/2\omega}$
Current = average emf / resistance

19. Using dimensional analysis or otherwise $t = \frac{V}{A}\sqrt{\frac{m}{RT}} \times$ some numerical factor
Substituting the values we get $\frac{t'}{t} = \frac{1}{2}\sqrt{\frac{1.7}{1.2}} \approx 0.6$

20. $\text{Force} = 25(0.4\hat{i} - 0.4\hat{k}) \times 2\hat{i} = -20\hat{j}$

21. The correct answer can be determined by dimensional analysis.

22. $\frac{\lambda D}{2d}$ = AB = 20 m
where λ = 2 m
D = 150 m
$\therefore$ d = 7.5 m

23. From geometry $\beta = (180° - 2\theta)$
So, $\frac{d\beta}{dt} = -\frac{2d\theta}{dt}$

24. When the switch closes at $V_C = 2V/3$ the capacitor discharges through $R_2 = 3R$
T(discharge) = 3RC ℓn 2 (since the voltage halves).
The charging occurs through $R_1 + R_2 = 9R$
T (charge) = 9RCℓn2
T = 12 RC ℓn 2 $\approx$ 8.4 RC

25. When the left end is positive, upper diode conducts and lower diode is cut off.
$R_{eq} = R$
When right end is positive the lower diode conducts.
$R_{eq} = R + \frac{3R}{4} = \frac{7R}{4}$
$\therefore$ Power = $\frac{(\Delta V)^2}{R}\left[1+\frac{4}{7}\right]\times\frac{1}{2} \approx \frac{(\Delta V)^2}{R}\times 0.8$

26. The object must be located at the centre of the curvature of the mirror for this to happen.
For the inverted image formed directly by the lens we can write
$\frac{1}{v}-\frac{1}{u}=\frac{1}{10}$, $\frac{v}{u}=-\frac{3}{2}$

27. This occurs when the angle of incidence is equal to the Brewster angle: tan $\theta_\beta = \mu$.

28. If the sphere has a uniform mass density (total mass m), then $\frac{\mu}{L}=\frac{q}{2m}$, where L = $\frac{2}{5}mR^2\omega$.

29. $h\nu = E_{n+1} - E_n = E_0\left(\frac{1}{n^2}-\frac{1}{(n+1)^2}\right) \approx \frac{E_0 2n}{n^4}$, for large n.

30. $m_n = \frac{m_1v_1 - m_2v_2}{v_2 - v_1}$
$= \frac{14\times(4.7\times10^6) - 1\times(3.3\times10^7)}{3.3\times10^7 - 4.7\times10^6} = 1.16$

Chemistry

PART – II

SECTION – A

31. $CaF_2 \rightleftharpoons \underset{s}{Ca^{2+}} + \underset{2s}{2F^-}$

$2s = 2\times10^{-4}$

$s = 10^{-4}$

$K_{sp} = 4s^3 = 4\times10^{-12}$

32. d-orbital is 5-fold degenerate in absence of magnetic field.

33. $\pi_{NaCl} = i\times CRT = 2\times0.1RT = 0.2\ RT$ $\quad \pi_{glucose} = 1\times0.1\times RT = 0.1\,RT$

$\pi_{NaCl} \neq \pi_{glucose}$

$\Delta T_{b(NaCl)} = 2\times0.1K_f$ $\quad \Delta T_{b(urea)} = 0.1\,K_f$

Hence $\Delta T_{b(urea)} < \Delta T_{b(NaCl)}$

$\Delta T_{f(glucose)} = 0.1\times K_f$ $\quad \Delta T_{f(KCl)} = 2\times0.1\times K_f$

$\Delta T_f = 0.2\,K_f$

$T_{f(glucose)} > T_{f(KCl)}$

34. $k = Ae^{-Ea/R\times600}$

$k^c = Ae^{-\frac{(Ea-20)}{R\times500}}$

$k = k^c$

$\frac{Ea}{R\times600} = \frac{(Ea-20)}{R\times500}$

$\frac{Ea}{6} = \frac{Ea-20}{5}$

$\frac{5}{6}Ea = Ea - 20$

$20 = Ea - \frac{5}{6}Ea$

$Ea = 20\times6$

$Ea = 120\ kJ/mol$

35. Melting point of ice decreases hence more amount of $H_2O(\ell)$ will form on increasing pressure.

36. $H^+(aq.) + OH^-(aq.) \longrightarrow H_2O(\ell)$

$\Delta_r H = \Delta_f H_{(H_2O)} - \left(\Delta_f H_{(H^+(aq.))} + \Delta_f H_{(OH^-)}\right)$

$-57.32 = -285.84 - \left(0 + \Delta_f H_{(OH^-(aq))}\right)$

$\Delta_f H_{(OH^-(aq))} = -228.52\ kJ/mol$

37. $4Fe + 10HNO_3 \longrightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$

38. $KE = \frac{1}{2}m \times v^2 = \frac{1}{2} \times 9 \times 10^{-31} \times \left(2.2 \times 10^6\right)^2$

$KE = \frac{1}{2} \times \frac{9 \times 10^{-31} \times \left(2.2 \times 10^6\right)^2}{1.6 \times 10^{-19}}$

$KE = 13.61\,eV$

$PE = -2 \times KE = -2 \times 13.61 = -27.22\ eV$

40. $\Lambda_M^o = \Lambda_N^o \times n - Factor$

$\Lambda_M^o(BaCl_2) = 160 \times 2 = 320$

$\Lambda_M^o(K_3PO_4) = 140 \times 3 = 420$

$\Lambda_M^o(KCl) = 100 \times 1 = 100$

$\Lambda^o_{M\left[Ba_3(PO_4)_2\right]} = 3\Lambda^o_{M(BaCl_2)} + 2\Lambda^o_{M(K_3PO_4)} - 6\Lambda^o_{M(HCl)}$

$= 3 \times 320 + 2 \times 420 - 6 \times 100$

= 1200

$\Lambda_M^o = \frac{K \times 1000}{s}$

$s = \frac{1.2 \times 10^{-5} \times 1000}{1200}$

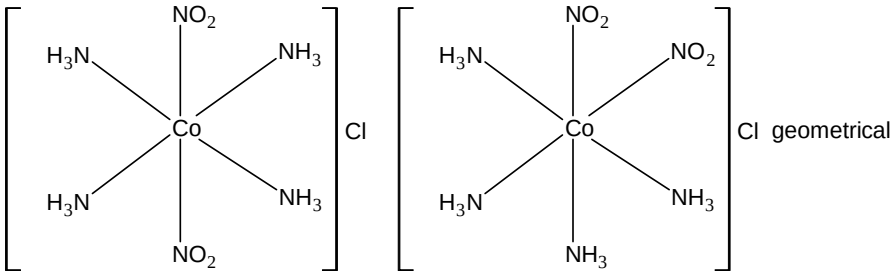
1×10^{-5}

$K_{sp\left[Ba_3(PO_4)_2\right]} = 108\ s^5$

$= 108 \times \left(10^{-5}\right)^5$

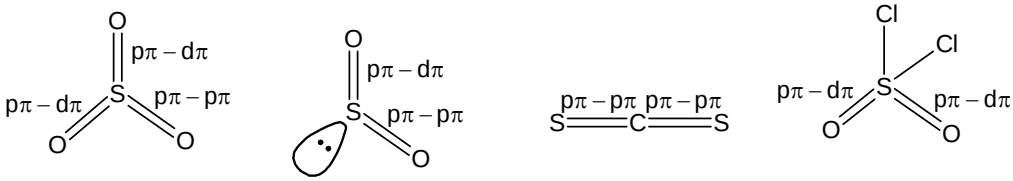
$= 1.08 \times 10^{-23}$

42. $[Co(NH_3)_4(ONO)_2]Cl$ $[Co(NH_3)_4(NO_2)_2]Cl$ linkage



$[Co(NH_3)_4(ONO)Cl]ONO$ $[Co(NH_3)_4(ONO)_2]Cl$ ionization.

43.

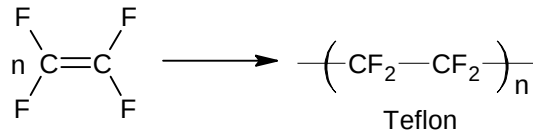


45. Pb^{2+} and Hg^{2+} both will precipitate out by H_2S in dil. HCl.

46. Oxidising power increases with increase of oxidation number of central atom.
Mn (O.S.) in MnO_4^- = +7
Cr (O.S.) in $Cr_2O_7^{2-}$ = +6
V (O.S.) in VO_2^+ = +5

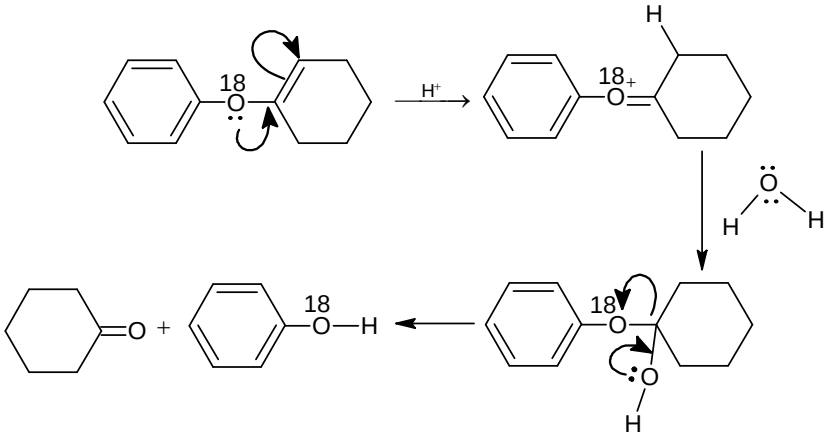
47. $Na_2S_2O_3 + Cl_2 + H_2O \longrightarrow NaHSO_4 + HCl$

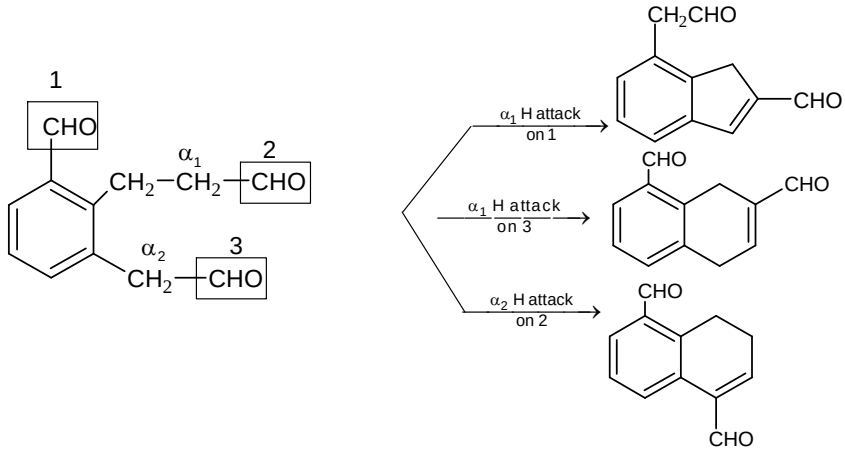
48. Ozone oxidizes alakaline solution of KI.

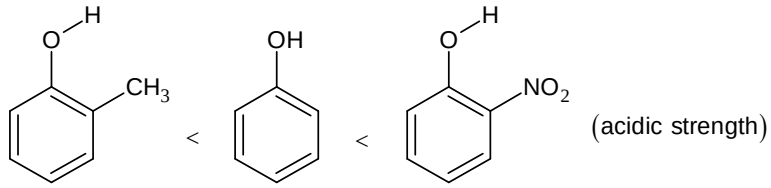
51. 

Teflon

52. Millon's test, Biuret test, Ninhydrin test are used for protein.

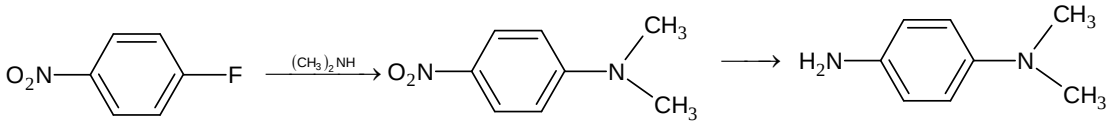
54. 

55. 

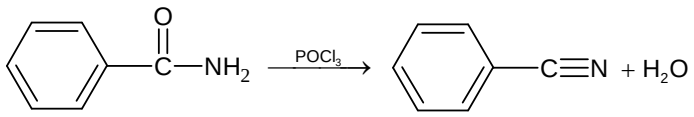
56. 

(acidic strength)

57.



58.



Mathematics

PART – III

SECTION – A

61. $\cot^{-1}\frac{5}{\sqrt{3}}+\cot^{-1}\frac{9}{\sqrt{3}}+\cot^{-1}\frac{15}{\sqrt{3}}+\cot^{-1}\frac{23}{\sqrt{3}}+.....\infty$

$= \tan^{-1}\left(\frac{\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{3}}}{1+\frac{2}{\sqrt{3}}\times\frac{1}{\sqrt{3}}}\right)+\tan^{-1}\left(\frac{\frac{3}{\sqrt{3}}-\frac{2}{\sqrt{3}}}{1+\frac{3}{\sqrt{3}}\times\frac{2}{\sqrt{3}}}\right)+.....+.....+\tan^{-1}\left(\frac{\frac{n+1}{\sqrt{3}}-\frac{n}{\sqrt{3}}}{1+\frac{n+1}{\sqrt{3}}\times\frac{n}{\sqrt{3}}}\right)+.....$

$\Rightarrow S_n=\tan^{-1}\frac{n+1}{\sqrt{3}}-\tan^{-1}\frac{1}{\sqrt{3}}$

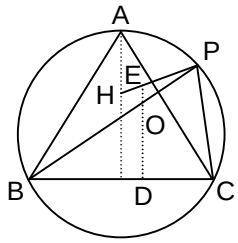
$\lim\limits_{n\to\infty} S_n=\frac{\pi}{2}-\frac{\pi}{6}=\frac{\pi}{3}$

62. Clearly tangents at P and Q intersect at right angle. Let S is point of intersection
$\Rightarrow$ P, Q, R, S are cyclic
$\Rightarrow$ S lies on the director circle of hyperbola
$\Rightarrow S\equiv\left(\sqrt{a^2-b^2}\cos\theta,\sqrt{a^2-b^2}\sin\theta\right)$

$\Rightarrow$ The chord with midpoint (h, k) i.e. circumcentre will be same as equation of the chord of contact w.r.t. S

$\Rightarrow \frac{xh}{a^2}-\frac{yk}{b^2}=\frac{h^2}{a^2}-\frac{k^2}{b^2}$ and $\frac{x\sqrt{a^2-b^2}\cos\theta}{a^2}-\frac{y\sqrt{a^2-b^2}\sin\theta}{b^2}=1$ are identical comparing and removing Q locus i $\frac{x^2+y^2}{a^2-b^2}=\left(\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)^2$

63. Let O (origin) is circumcentre of ΔABC and position vectors of A, B, C and P are $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{p}$



Hence $\overrightarrow{OH}=\vec{a}+\vec{b}+\vec{c}$, $\overrightarrow{OD}=\frac{\vec{b}+\vec{c}}{2}$, $\overrightarrow{OE}=\frac{\vec{a}+\vec{b}+\vec{c}+\vec{p}}{2}$

$\overrightarrow{DE}=\overrightarrow{OE}-\overrightarrow{OD}=\frac{\vec{a}+\vec{p}}{2}$, $\overrightarrow{AP}=\vec{p}-\vec{a}$

$\Rightarrow \overrightarrow{DE}\cdot\overrightarrow{AP}=\frac{|\vec{p}|^2-|\vec{a}|^2}{2}=0$ since $|\vec{p}|=|\vec{a}|$

64. It is equal to $\int_0^1\left(\frac{x^3}{4}-x\left(f(x)-\frac{x}{2}\right)^2\right)dx$ which is less than $\int_0^1\frac{x^3}{4}dx=\frac{1}{16}$

65. $f(x)=a_nx^n+a_{n-1}x^{n-1}+a_{n-2}x^{n-2}+.....+a_0$, $[x]=m$ then $f(m)=a_nm^n+a_{n-1}m^{n-1}+.....+a_0$
$f(f(m)+1)=a_n(f(m)+1)^n+a_{n-1}(f(m)+1)^{n-1}+.....+a_0$
$kf(m)+a_n+a_{n-1}+.....+a_0=kf(m)+f(1)$

Since $\frac{f(f(m)+1)}{f(m)}=k+\frac{f(1)}{f(m)}$ = integer

$\Rightarrow m=1$ hence $x\in[1,2)$

66. Total such numbers are $(3 \times 2) \times (3!) = 36$
Sum of all such numbers $66666 \times 18 = 1199988$

67. $(z + \omega)^2 = z\omega \Rightarrow z^2 + \omega^2 + z\omega = 0$
$\Rightarrow \frac{z}{\omega} = \frac{-1 \pm \sqrt{3}\,i}{2} \Rightarrow \arg(z) - \arg(\omega) = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$

68. $\left|\vec{p} \times \vec{q} + \vec{q} \times \vec{r} + \vec{r} \times \vec{p}\right| = \left|(\vec{q} - \vec{p}) \times (\vec{r} - \vec{p})\right| = \left|\vec{q} - \vec{p}\right|\left|\vec{r} - \vec{p}\right| \sin 60^o = \frac{a^2\sqrt{3}}{2}$

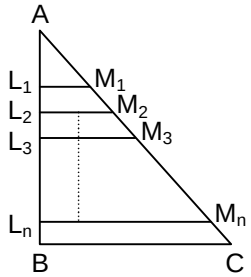
69. $(x + y + z)^6 \rightarrow$ general term $\frac{\lfloor 6}{\lfloor \alpha \lfloor \beta \lfloor \gamma} x^\alpha y^\beta z^\gamma$
$\alpha + \beta + \gamma = 6$, $\alpha \neq 0$ then total number of solution $= {}^8C_2 - 7 = 21$

70. $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot b & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot a & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot b & \vec{c}\cdot\vec{c} \end{vmatrix}$

71. $1 + \omega + \omega^2 = 0$ and $\begin{vmatrix} \frac{-x}{\omega^2} & -y & \frac{-z}{\omega} \\ -y & \frac{-z}{\omega} & \frac{-x}{\omega^2} \\ \frac{-z}{\omega} & \frac{-x}{\omega^2} & -y \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz$
$= (x + y + z)(x + y\omega + z\omega^2)(x + y\omega^2 + z\omega) = 0$
$= (x + y + z)((x - y)^2 + (y - z)^2 + (z - x)^2) = 0$
$\Rightarrow$ Either $x = y = z$ or $x + y\omega + z\omega^2 = 0$

72. Denominator of all terms equal to $(2a_{n+1})$
Numerator $= (2nd + (2n - 2)d + ... + 2d) = \frac{2(n(n+1))}{2}d$

73. $\frac{AL_1}{AB} = \frac{L_1M_1}{BC}$
$\Rightarrow \frac{1}{n+1} = \frac{L_1M_1}{a} \Rightarrow L_1M_1 = \frac{a}{n+1}$
$\frac{AL_2}{AB} = \frac{L_2M_2}{BC} \Rightarrow \frac{2}{n+1} = \frac{L_2M_2}{a} \Rightarrow L_2M_2 = \frac{2a}{n+1}$
$\Rightarrow$ The required sum is $\frac{a}{n+1} + \frac{2a}{n+1} + \frac{3a}{n+1} + ... + \frac{na}{n+1} = \frac{an}{2}$



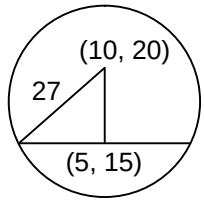
74. ${}^nC_0\left(\frac{1}{2} - \frac{1}{3}\right) - {}^nC_1\left(\frac{1}{3} - \frac{1}{4}\right) + {}^nC_2\left(\frac{1}{4} - \frac{1}{5}\right) - ..... + {}^nC_n(-1)^n\left(\frac{1}{n+2} - \frac{1}{n+3}\right)$
$\int_0^1 (1-x)^n x\,dx = \int_0^1 \left({}^nC_0 x - {}^nC_1 x^2 + {}^nC_2 x^3 + ... + (-1)^n\ {}^nC_n x^{n+1}\right)dx$

$$\int_0^1 (1-x)^n x^2 dx = \int_0^1 \left({}^nC_0 x^2 - {}^nC_1 x^3 + {}^nC_2 x^4 + ... + (-1)^n {}^nC_n x^{n+2} \right) dx$$

$$= \frac{1}{(n+1)(n+2)} - \frac{2}{(n+2)(n+1)(n+3)} = \frac{1}{(n+1)(n+2)}\left[1 - \frac{2}{n+3}\right) = \frac{1}{(n+2)(n+3)}$$

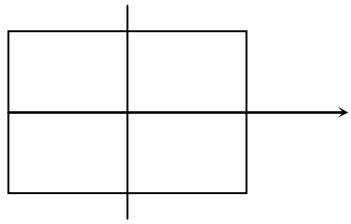
75. Coefficient of x^n in $(x + x^2 .....)(x^2 + x^4 + x^6 .....)(x^3 + x^6 + x^9 + ...)$
$= x^6(1 - x)^{-1}(1 - x^2)^{-1}(1 - x^3)^{-1}$
$\Rightarrow$ None of the option can satisfy

76. Clearly (5, 15) is the mid point of the chord hence slope of normal to the chord is $\frac{20-15}{10-5} = 1$, slope of chord = –1



Equation of chord $(y - 15) = -(x - 5)$
$\Rightarrow x + y = 20$
$\Rightarrow 2x + 2y = 40$
$\Rightarrow p^{2013} + q^{2013} = (2)^{2013} + (2)^{2013} = 2^{2014}$

77. Let origin is shifted to (1, 0) after this the equation will be $\max\{|x|, |y|\} = k$
Area $= 4k^2 = 100$
$\Rightarrow k = 5$



78. $$\lim_{x\to\infty} x^2 \ell n\left(x \cot^{-1} x\right) = \lim_{h\to 0} \ell n\left(\frac{\tan^{-1} h}{h}\right)^{\frac{1}{h^2}} = \lim_{h\to 0} \frac{1}{h^2}\left(\frac{\tan^{-1} h}{h} - 1\right) = \lim_{h\to 0} \frac{\left(h - \frac{h^3}{3} + \frac{h^5}{5} .....\right) - h}{h^3} = \frac{1}{3}$$

79. $$I = \int \frac{\cos x + \sqrt{3}}{\left(1 + 2\sin\left(x + \frac{\pi}{3}\right)\right)^2} dx = \int \frac{\cos x + \sqrt{3}}{\left(1 + \sin x + \sqrt{3}\cos x\right)^2} dx$$

Dividing by $\sin^2 x$, we get $I = \int \frac{\text{cosec}\, x \cot x + \sqrt{3}\ \text{cosec}^2 x}{\left(\text{cosec}\, x + 1 + \sqrt{3}\cot x\right)^2} dx = \frac{1}{\text{cosec}\, x + 1 + \sqrt{3}\cot x} + c$

$$= \frac{\sin x}{1 + 2\sin\left(x + \frac{\pi}{3}\right)} + c$$

80. $$f(x) = \frac{1}{2\{-x\}} - \left(1 - \{-x\}\right) = \{-x\} + \frac{1}{2\{-x\}} - 1$$

$\Rightarrow f(x) \geq 2\sqrt{\frac{1}{2}} - 1$ (A.M. $\geq$ G.M.)

$\Rightarrow f_{min} = \sqrt{2} - 1$ which occurs when $\{-x\} = \frac{1}{\sqrt{2}}$

$\Rightarrow -x = n + \frac{1}{\sqrt{2}} \Rightarrow n - \frac{1}{\sqrt{2}}, n \in I$

81. $$\int\frac{x^3-x^2+x-1}{x^5+1}dx=\int\frac{x^4-1}{\left(x^5+1\right)\left(x+1\right)}dx=\int\frac{x^4}{x^5+1}dx-\int\frac{1}{x+1}dx$$
$$=\frac{1}{5}\ell n\left|x^5+1\right|-\ell n\left|x+1\right|+c$$

82. Odd power of a skew symmetric matrix is always a skew symmetric matrix

83. $F \cap G = \{3, 6\}$, $E \cup \{F \cap G\} = \{1, 3, 6, 5, 7\}$
$\Rightarrow P\{E \cup \{F \cap G)\} = 0.13 + 0.12 + 0.13 + 0.08 + 0.06 = 0.52$

84. Plane containing the line will be given as $(4x – 2y – 10) + \lambda(5y – 4z + 3) = 0$
$\Rightarrow 4x + (5\lambda – 2)y + 4\lambda z + 3\lambda – 10 = 0$
The plane passing through (4, 3, 7) will give λ
$\Rightarrow 16 + (5\lambda – 2)y + 4\lambda z + 3\lambda – 10 = 0$
$\Rightarrow –10\lambda = 0 \Rightarrow \lambda = 0$
i.e., $4x – 2y – 10 = 0$

85. Let $f(x) = \sin^4 x – \sin x \cos x + \cos^4 x = 1-\frac{1}{2}\sin 2x-\frac{1}{2}\sin^2 2x$
Let $\sin 2x = t$, $f(x) = g(t) \Rightarrow g(t) = \frac{-1}{2}t^2-\frac{1}{2}t+1$
Since $t \in [–1, 1] \Rightarrow f\left(x\right)\in\left[0,\frac{9}{8}\right]$

86. $f\left(x\right)=\sqrt{3-x}-\sqrt{x+1}$ is well defined for $x \in [–1, 3]$
$f'\left(x\right)=\frac{-1}{2\sqrt{3-x}}-\frac{1}{2\sqrt{x+1}}<0 \ \forall\ x\in R$
$f(–1) = 2$ and $\sqrt{12-4x}=1+\sqrt{4x-4}$ gives $x=1-\frac{\sqrt{31}}{8}$
Hence, $\left[x\in\left[-1,1-\frac{\sqrt{31}}{8}\right]\right)$

87. $\alpha^n = 1$, $\alpha^{n+1} = \alpha$ ..... etc.
$\left(\alpha^P\right)^{n/P}=1$ is P is divisor of n

88. ΔABC is isosceles and circumcentre of ΔPQR is orthocentre of ΔABC and lies on a line $\perp$ to $x – 3y – 31 = 0$ passing through (0, 3)

89. Roots of the equation $x^2 + kx + 2$ are irrational since $k^2 – 8$ is not a perfect square as $k > 12$
So $\frac{a}{1}=\frac{c}{2}=\frac{\sin C+\sin A}{\sin C-\sin A}=3$

90. At point of maxima $f'(x) = 0$ and $f''(x) < 0$
$\Rightarrow f''(x) = x^2 – f^2(x) \le 0$
Since the curve $x^2 – y^2 = a^2$ and $x^2 – f^2(x) \le 0$
$x_1^2-y_1^2<a^2 \Rightarrow$ Point lies outside hyperbola
Hence, 2 tangents